

## Chapter 5: Functions – Practice Examples V1

### 1. Calculate Price with VAT:

Write a function called `calculate_vat` that takes a product price as a parameter and returns the total price after adding 15% VAT.

### 2. Convert Celsius to Fahrenheit:

Create a function named `celsius_to_fahrenheit` that takes a temperature in Celsius and returns the temperature in Fahrenheit using this formula:  $F = (C \times 9/5) + 32$

### 3. Generate Email Address:

Write a function called `create_email` that takes a name and a surname as parameters and returns an email address in the format:

`[name].[surname]@nwu.com`

*Example: If name is John and surname is Smith, return "john.smith@nwu.com".*

### 4. Convert Hours to Minutes:

Write a function named `hours_to_minutes` that takes a number of hours as input and returns the equivalent number of minutes.

### 5. Get Grade Symbol:

Write a function called `get_grade` that takes a percentage mark and returns a grade symbol:

- Return "A" if 75 or above
- "B" if 60–74
- "C" if 50–59
- "F" if below 50

### 6. Calculate Average Speed:

Create a function called `average_speed` that takes distance (in kilometres) and time (in hours), and returns the speed in km/h.

## 7. Area of a circle:

Write a function, with an appropriate name, that takes in the radius as a parameter and returns the area of a circle. Make use of the math module for Pi. Round to 2 integers using the round() function.

## 8. Random Guessing Game:

Create a program that lets the user enter a number to see whether they guessed the correct number between 1 and 50. Each task should be its own function. Ask the user to repeat until they have the correct number and return the number of tries it took.

## 9. Find the Higher Number:

Write a function called `get_max` that takes two numbers and returns the higher one

## 10. Recursive Factorial Calculator (Homework – Very Advanced!):

Write a function named `factorial` that uses recursion to calculate the factorial of a given positive integer.

The factorial of a number  $n$  is the product of all positive integers from 1 to  $n$ .

### Definition:

- `factorial(1)` should return 1 (this is the base case)
- For all  $n > 1$ : `factorial(n)` should return  $n * \text{factorial}(n - 1)$

### Example:

- `factorial(4)` returns 24 → because  $4 \times 3 \times 2 \times 1 = 24$

Only use return statements, not `print()` inside the function. Display the result in `main()`.